Enhancing Reasoning for Diffusion LLMs via Distribution Matching Policy Optimization

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Diffusion LLMs (dLLMs) & Reinforcement Learning (RL)

■ **Goal**: Improve reasoning of dLLMs with an RL method tailored to masked diffusion structure.

Masked diffusion models learn the conditional distribution: input a partially masked sequence $\boldsymbol{x} = (x_1, \dots, x_D)$, output $\boldsymbol{\pi}_\theta(\boldsymbol{x}) \in \mathbb{R}^{D\times V}$, where

$$\boldsymbol{\pi}_\theta(\boldsymbol{x})_{d,u} \approx \Pr_{\boldsymbol{X}\sim p_{\text{data}}}(X_d = u | \boldsymbol{X}_{\text{UM}} = \boldsymbol{x}_{\text{UM}}), \text{ if } x_d = \mathsf{M}.$$

Sequence probability defined by **random-order autoregressive** generation:

$$p_\theta(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{\sigma}}\, p_\theta(\boldsymbol{x}; \boldsymbol{\sigma}), \text{ where } p_\theta(\boldsymbol{x}; \boldsymbol{\sigma}) = \prod_{d=1}^{|\boldsymbol{x}|} \pi_\theta(x_{\sigma_d} | \boldsymbol{x}_{\sigma_{<d}}).$$

The **ELBO** is a surrogate of log-probability that leads to the denoising cross-entropy loss $\min_\theta \mathbb{E}_{p_{\text{data}}(\boldsymbol{x})}\, \mathcal{L}_\theta(\boldsymbol{x})$:

$$\begin{aligned} -\log p_\theta(\boldsymbol{x}) &\le -\mathbb{E}_{\boldsymbol{\sigma}} \log p_\theta(\boldsymbol{x}; \boldsymbol{\sigma}) \\ &= \mathop{\mathbb{E}}_{m\sim\text{Unif}\{1,\dots,|\boldsymbol{x}|\}} \left[\frac{|\boldsymbol{x}|}{m} \mathbb{E}_{\mu_m(\widetilde{\boldsymbol{x}}|\boldsymbol{x})} \sum_{d:\widetilde{x}_d = \mathsf{M}} -\log \pi_\theta(\widetilde{\boldsymbol{x}})_{d,x_d} \right] =: \mathcal{L}_\theta(\boldsymbol{x}). \end{aligned}$$

Here $\mu_m(\cdot|\boldsymbol{x})$ samples a uniformly random subset of $\{1, \dots, |\boldsymbol{x}|\}$ of size m and masks the corresponding entries. For reasoning, we write $\boldsymbol{x} = (\boldsymbol{q}, \boldsymbol{o})$ (prompt, response).

■ **Why existing RL methods for LLMs are suboptimal for dLLMs?**

- Sequence likelihood is expensive/inexact for bidirectional generation.
- GRPO-style methods are mostly backward-trajectory based.
- Reward maximization alone tends to be mode-seeking.

Policy Distribution Matching Learning. Given a pretrained dLLM policy $\boldsymbol{\pi}_{\text{ref}}(\boldsymbol{o}|\boldsymbol{q})$ that samples from $p_{\text{ref}}(\boldsymbol{o}|\boldsymbol{q})$, a reward function $r : (\boldsymbol{q}, \boldsymbol{o}) \mapsto \mathbb{R}$, a set of prompts $\mathcal{D}$, and temperature $\alpha > 0$, learn a dLLM policy $\boldsymbol{\pi}_\theta(\boldsymbol{o}|\boldsymbol{q})$ that produces the desired optimal distribution

$$p_*(\boldsymbol{o}|\boldsymbol{q}) \propto p_{\text{ref}}(\boldsymbol{o}|\boldsymbol{q}) \exp\left(\frac{r(\boldsymbol{q}, \boldsymbol{o})}{\alpha}\right) \text{ via } \min_{\boldsymbol{\pi}_\theta} \mathbb{E}_{\boldsymbol{q}\sim\mathcal{D}}\, \mathcal{F}(p_\theta(\cdot|\boldsymbol{q}), p_*(\cdot|\boldsymbol{q})).$$

References

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Distribution Matching Policy Optimization (DMPO)

▶ Core loss: Weighted Denoising Cross-Entropy (WDCE) [5]:

$$\min_\theta \mathop{\mathbb{E}}_{\boldsymbol{q}\sim\mathcal{D}} \mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \mathop{\mathbb{E}}_{p_{\text{old}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})} \left\{ w(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma}) \mathop{\mathbb{E}}_{m\sim\text{Unif}\{1,\dots,|\boldsymbol{o}|\}} \left[\frac{|\boldsymbol{o}|}{m} \mathop{\mathbb{E}}_{\mu_m(\widetilde{\boldsymbol{o}}|\boldsymbol{o})} \sum_{d:\widetilde{o}_d = \mathsf{M}} -\log \boldsymbol{\pi}_\theta(\widetilde{\boldsymbol{o}}|\boldsymbol{q})_{d,o_d} \right] \right\}.$$

Treat i.i.d. samples from the old policy p_{old} as weighted samples from the optimal p_*: compute the weights by

$$w(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma}) = \frac{p_*(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})}{p_{\text{old}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})} \propto \exp\left(\frac{r(\boldsymbol{q},\boldsymbol{o})}{\alpha} + \log\frac{p_{\text{ref}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})}{p_{\text{old}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})}\right) =: e^{\ell(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})},$$

followed by softmax normalization over a group of rollouts for the same prompt.

▶ **Small-batch issue & fix.** All-positive weights may promote both good and bad responses when rollouts per prompt are limited. We insert **negative gradients** by baseline subtraction:

$$w_{\text{real}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma}) = w(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma}) - w_{\text{base}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma}), \quad \text{e.g., } w_{\text{base}} \equiv 1.$$

▶ Besides WDCE (forward-KL style), we also explore **weighted direct discriminative optimization (WDDO)** [4]:

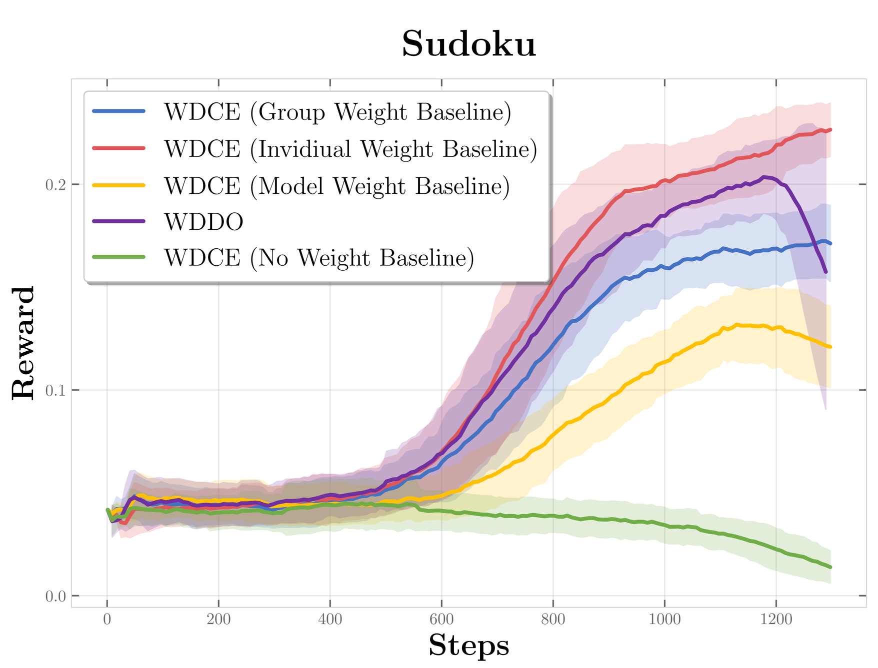
$$\mathcal{F} = -\mathop{\mathbb{E}}_{\boldsymbol{\sigma}} \mathop{\mathbb{E}}_{p_{\text{old}}(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma})} \left[w(\boldsymbol{o}|\boldsymbol{q};\boldsymbol{\sigma}) \log\sigma\left(\log\frac{p_\theta(\boldsymbol{o}|\boldsymbol{q})}{p_{\text{old}}(\boldsymbol{o}|\boldsymbol{q})}\right) + \log\sigma\left(-\log\frac{p_\theta(\boldsymbol{o}|\boldsymbol{q})}{p_{\text{old}}(\boldsymbol{o}|\boldsymbol{q})}\right)\right],$$

which naturally introduces positive/negative gradient competition and shares the same optimum p_*.

▶ **Practical properties of DMPO**

- Off-policy: supports replay buffer and stale roll-out reuse.
- Forward-only: training leverages cheap noising, not full trajectories.
- Compatible with fast-dLLM samplers [2] for additional speed-up.

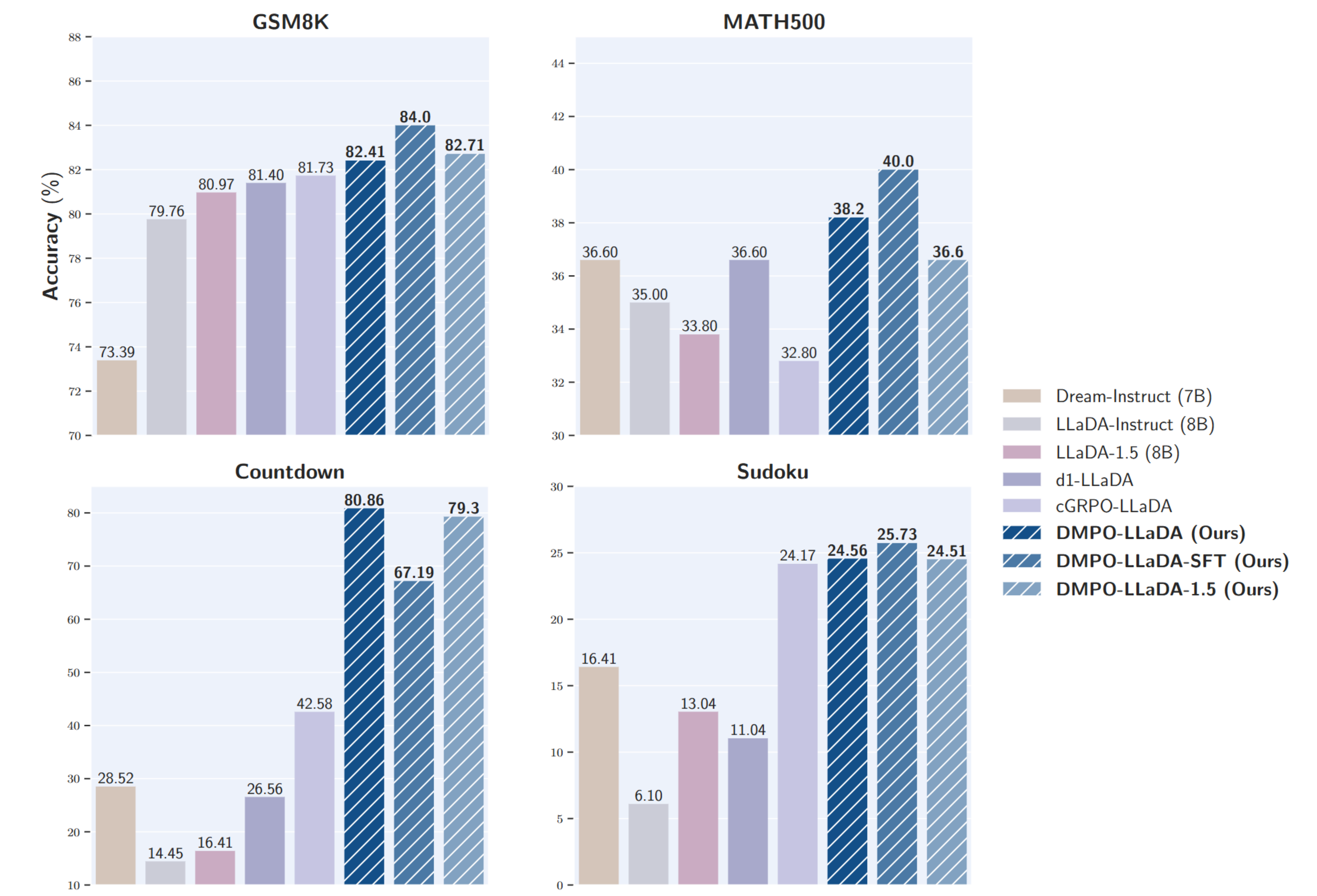
▶ **Ablation: baseline subtraction is crucial** for stable learning at small batch size.



Takeaway. DMPO turns dLLM RL from reward-mode seeking into *distribution matching*, via off-policy and forward-only training tailored to dLLMs.

Empirical Results: Strong Reasoning Gains & Efficient Training

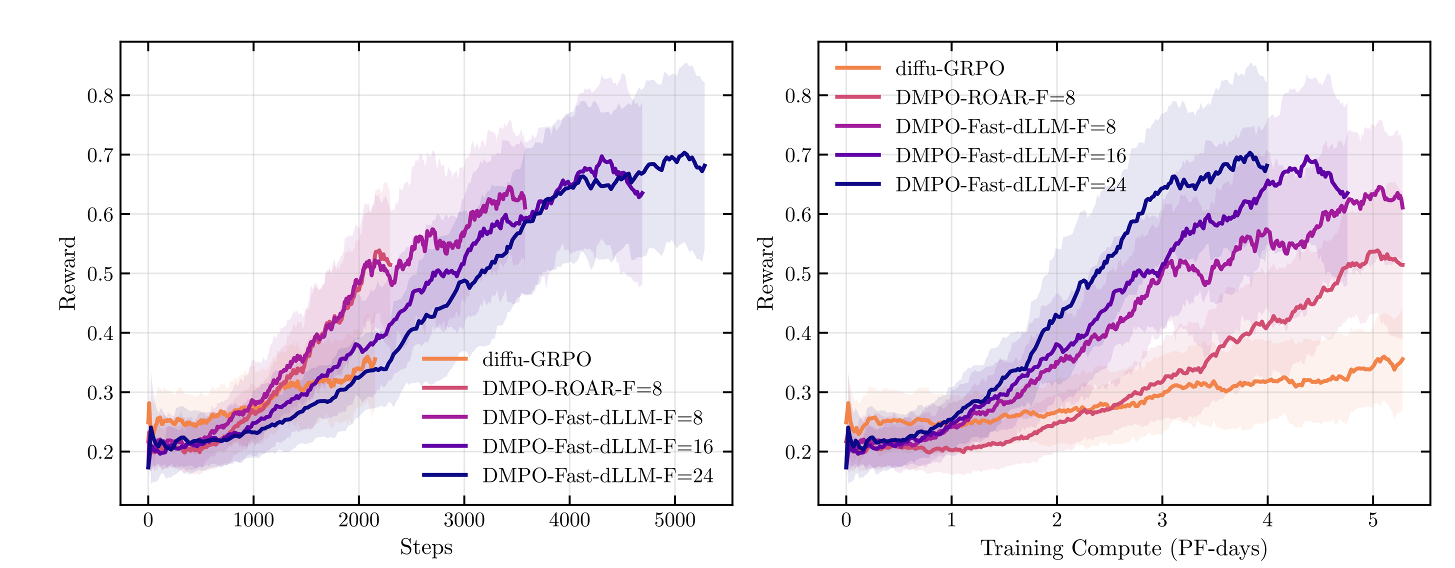
⧫ **Main benchmark on LLaDA-Instruct (8B).** DMPO consistently outperforms dLLM baselines on GSM8K, MATH500, Countdown, and Sudoku; gains over d1 [3] & cGRPO [1] are especially large on planning-heavy tasks.



⧫ DMPO generalizes to a different backbone: Dream-Instruct (7B)

Model	GSM8K		MATH500		Countdown		Sudoku	
Gen. length	256	512	256	512	256	512	256	512
Dream-Instruct	73.39	76.65	36.60	36.40	28.52	27.34	16.41	11.77
d1-Dream	80.52	82.41	41.20	45.60	29.46	36.83	21.78	23.34
DMPO-Dream (Ours)	84.03	84.76	47.40	47.20	54.43	56.51	41.04	45.55

⧫ **Training efficiency.** DMPO is **sample-efficient** (buffer reuse via off-policy weights) and **compute-efficient** (benefits from fast-dLLM [2] sampling).



Paper link on ArXiv: https://arxiv.org/abs/2510.08233

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